This project aims to develop a machine learning model that predicts lithology (either reservoir rock or non-reservoir rock) from well logs using Gamma ray logs, DPHI porosity, Neutron porosity, caliper logs, and Deep Induction log

Real well log data was obtained from Kansas field and available at https://kgsimages.blob.core.windows.net/web/web_1/WebDocs/WellLogs/kcc

In [51]:

```
# Install the needed libraries
import lasio
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report
```

1. Read the well log data from the webpage and clean it up by dropping nulls

In [84]:

```
las = lasio.read('https://kgsimages.blob.core.windows.net/web/web_1/WebDocs/WellLogs/kcc
df = las.df()
df.head()
```

Out[84]:

	GR	CALY	CALM	CALX	SP	TENS	BS	SFL	ILD	ILM	DPHI	NPHI	PE	MINV	MN(
DEPT															
244.0	69.678	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
244.5	71.686	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
245.0	75.918	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
245.5	80.436	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
246.0	76.796	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N

In [52]:

```
las = lasio.read('Log data Kansas.las')
df = las.df()
df.head()
```

Out[52]:

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DEPT	GR	CALY	CALM	CALX	SP	TENS	BS	SFL	ILD	ILM	DPHI	NPHI	PE	MINV	MNC
244.0	69.678	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
244.5	71.686	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
245.0	75.918	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
245.5	80.436	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
246.0	76.796	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N

In [53]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 6073 entries, 244.0 to 3280.0
Data columns (total 18 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   GR      5941 non-null   float64
 1   CALY    5805 non-null   float64
 2   CALM    5810 non-null   float64
 3   CALX    5816 non-null   float64
 4   SP      5943 non-null   float64
 5   TENS    5942 non-null   float64
 6   BS      5918 non-null   float64
 7   SFL     5938 non-null   float64
 8   ILD     5944 non-null   float64
 9   ILM     5916 non-null   float64
 10  DPHI    5950 non-null   float64
 11  NPHI    5818 non-null   float64
 12  PE      5948 non-null   float64
 13  MINV    5936 non-null   float64
 14  MNOR    5936 non-null   float64
 15  DRHO    5946 non-null   float64
 16  DT      5946 non-null   float64
 17  SPHI    5946 non-null   float64
dtypes: float64(18)
memory usage: 901.5 KB
```

In [54]:

```
# the nulls will be dropped and not interpolated
df = df.dropna()
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 5750 entries, 339.5 to 3214.0
Data columns (total 18 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   GR      5750 non-null   float64
 1   CALY    5750 non-null   float64
 2   CALM    5750 non-null   float64
 3   CALX    5750 non-null   float64
 4   SP      5750 non-null   float64
 5   TENS    5750 non-null   float64
 6   BS      5750 non-null   float64
 7   SFL     5750 non-null   float64
```

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```
 8   ILD      5750 non-null   float64
 9   ILM      5750 non-null   float64
 10  DPHI     5750 non-null   float64
 11  NPHI     5750 non-null   float64
 12  PE       5750 non-null   float64
 13  MINV     5750 non-null   float64
 14  MNOR     5750 non-null   float64
 15  DRHO     5750 non-null   float64
 16  DT       5750 non-null   float64
 17  SPHI     5750 non-null   float64
dtypes: float64(18)
memory usage: 853.5 KB
```

In [55]:

```
df.head()
```

Out[55]:

	GR	CALY	CALM	CALX	SP	TENS	BS	SFL	ILD	ILM	DPHI	NPHI
DEPT												
339.5	97.759	8.915	8.112	9.482	33.252	1792.279	7.873	30.719	24.408	25.642	19.270	29.258
340.0	107.505	8.943	8.186	9.527	33.485	1813.362	7.873	28.734	22.865	23.331	18.986	29.006
340.5	115.464	8.938	8.220	9.536	34.530	1825.388	7.873	28.568	21.419	21.320	19.554	29.977
341.0	119.610	8.929	8.226	9.574	35.577	1829.193	7.873	27.998	20.066	19.482	19.866	29.971
341.5	127.681	8.920	8.231	9.633	36.481	1832.998	7.873	25.849	18.798	17.801	20.276	29.965

2. Create the dependent variable (lithology) from gamma ray values

Using the feature similar to IF in Ms Excel, if GR < 75 (False = 0, True = 1) Implying 0 = True = Sand 1 = False = Shale

In [56]:

```
df['Lithology'] = np.where (df['GR'] < 75, 0, 1)
df['Lithology'].head()
```

Out[56]:

```
DEPT
339.5    1
340.0    1
340.5    1
341.0    1
341.5    1
Name: Lithology, dtype: int32
```

In [57]:

```
y = df['Lithology']
```

3. List and create the independent variable for the model

Gamma Ray, Caliper Log, Density porosity, and deep induction logs wil be used for predictig the lithology

In [58]:

```
df.columns
```

Out[58]:

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```
Index(['GR', 'CALY', 'CALM', 'CALX', 'SP', 'TENS', 'BS', 'SFL', 'ILD', 'ILM',
       'DPHI', 'NPHI', 'PE', 'MINV', 'MNOR', 'DRHO', 'DT', 'SPHI',
       'Lithology'],
      dtype='object')
```

In [59]:

```
Features = ['GR', 'CALX', 'ILD', 'DPHI', 'NPHI']
X = df[Features]
```

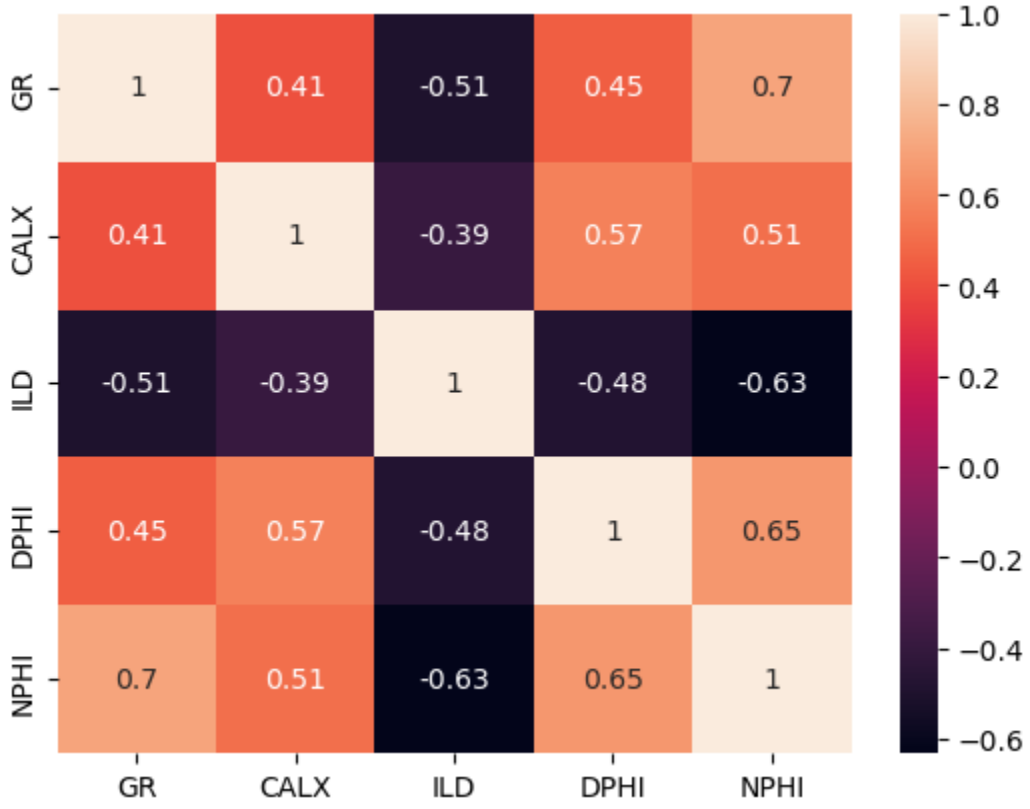
3a. Check for collinearity between the independent variavles before proceeding

In [60]:

```
corrr = X.corr()
sns.heatmap(corrr, annot = True)
corrr
```

Out[60]:

	GR	CALX	ILD	DPHI	NPHI
GR	1.000000	0.406116	-0.505212	0.446537	0.701930
CALX	0.406116	1.000000	-0.391660	0.567353	0.506967
ILD	-0.505212	-0.391660	1.000000	-0.484239	-0.633122
DPHI	0.446537	0.567353	-0.484239	1.000000	0.651580
NPHI	0.701930	0.506967	-0.633122	0.651580	1.000000



From the above correlation matrix, the porosities (DPHI, NPHI) and induction logs appear to have a moderate relationship with one another. However, we will proceed since such a relationship is not unexpected.

4. Data Splitting and model training

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In [61]:

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state
model = RandomForestClassifier(n_estimators = 100)
model.fit(X_train, y_train)
```

Out[61]:

```
▾ RandomForestClassifier
RandomForestClassifier()
```

5. Use the model to predict and evaluate the pedictions

In [62]:

```
y_pred = model.predict(X_test)
print (classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       537
           1       1.00      1.00      1.00       613

    accuracy                           1.00      1150
   macro avg       1.00      1.00      1.00      1150
weighted avg       1.00      1.00      1.00      1150
```

6. Interpretation of model prediction result

The classification report evaluates the performance of the machine learning model by comparing predicted lithology against the true labels. It reports metrics such as precision, recall, and F1-score for each class. In my case the model achieved perfect accuracy, which likely occurs because the lithology labels were derived from the gamma ray log, which is also included as an input feature.

7. Determining the importance of each feature in predicting lithology

In [63]:

```
Importance = model.feature_importances_
plt.figure(figsize =(8,6))
plt.bar(Features, Importance)
plt.title("Feature Importance for Lithology Prediction")
plt.xlabel('Feature List')
plt.ylabel('Degree of Importance in Predicitng Lithology')
plt.show()
```

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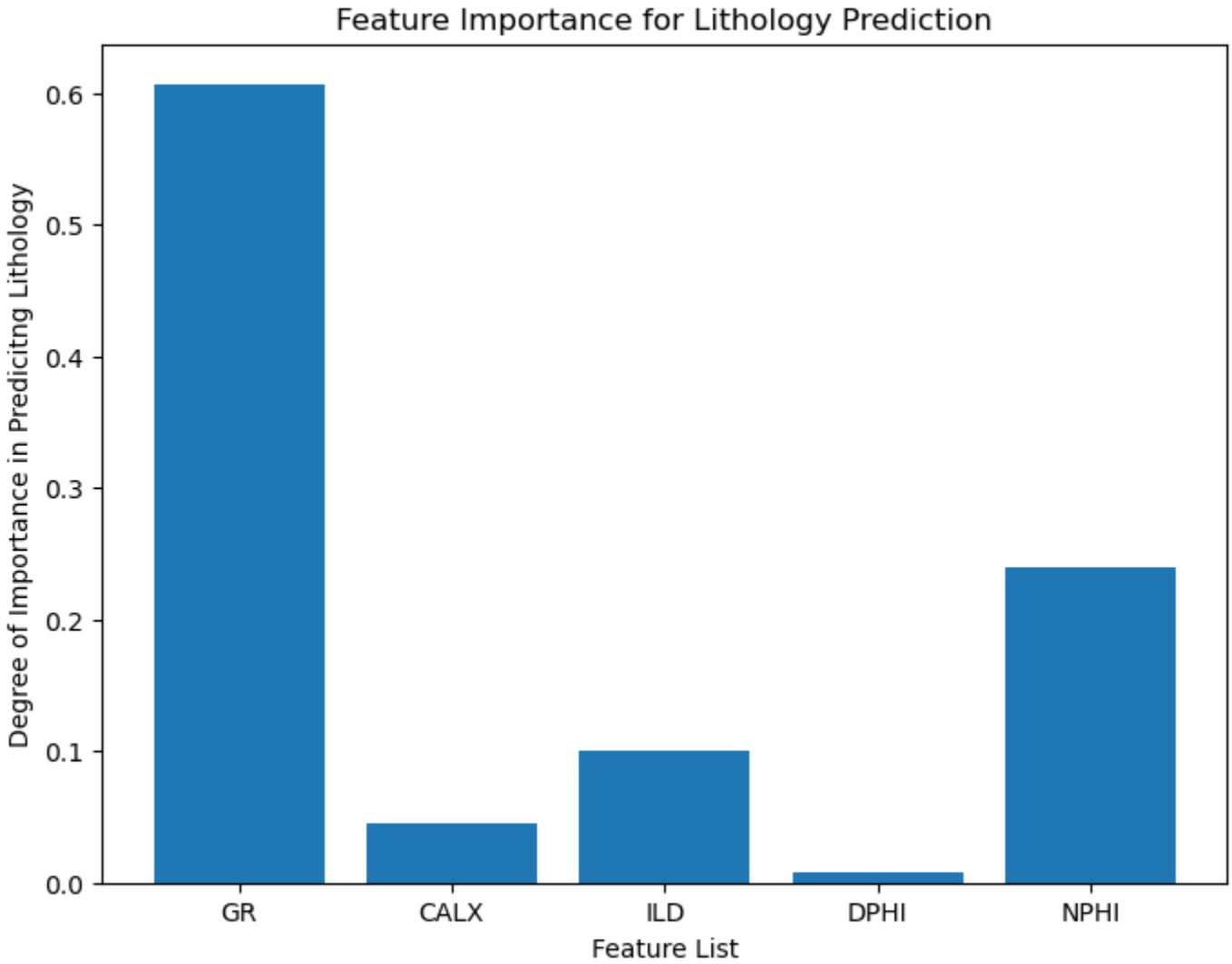

The above graph reveals that gamma ray is the most significant predictor of the lithology type. This is not unexpected due to the inherent use of gamma ray values to differentiate between lithologies. The second most importnt determinant of the lithology type is neutron porosity and then followed by deep induction logging values of the interval.

8. Plot the lithology for the entire logged interval using the developed model

In [75]:

```
df['Predicted_lithology'] = model.predict(X)
plt.figure(figsize=(4,8))
plt.scatter(df['Predicted_lithology'], df.index, s=2)
plt.ylabel('Depth')
plt.xlabel('Predicted Lithology')
plt.key(
plt.gca().invert_yaxis()

plt.title("Predicted Lithology")

plt.show()
```

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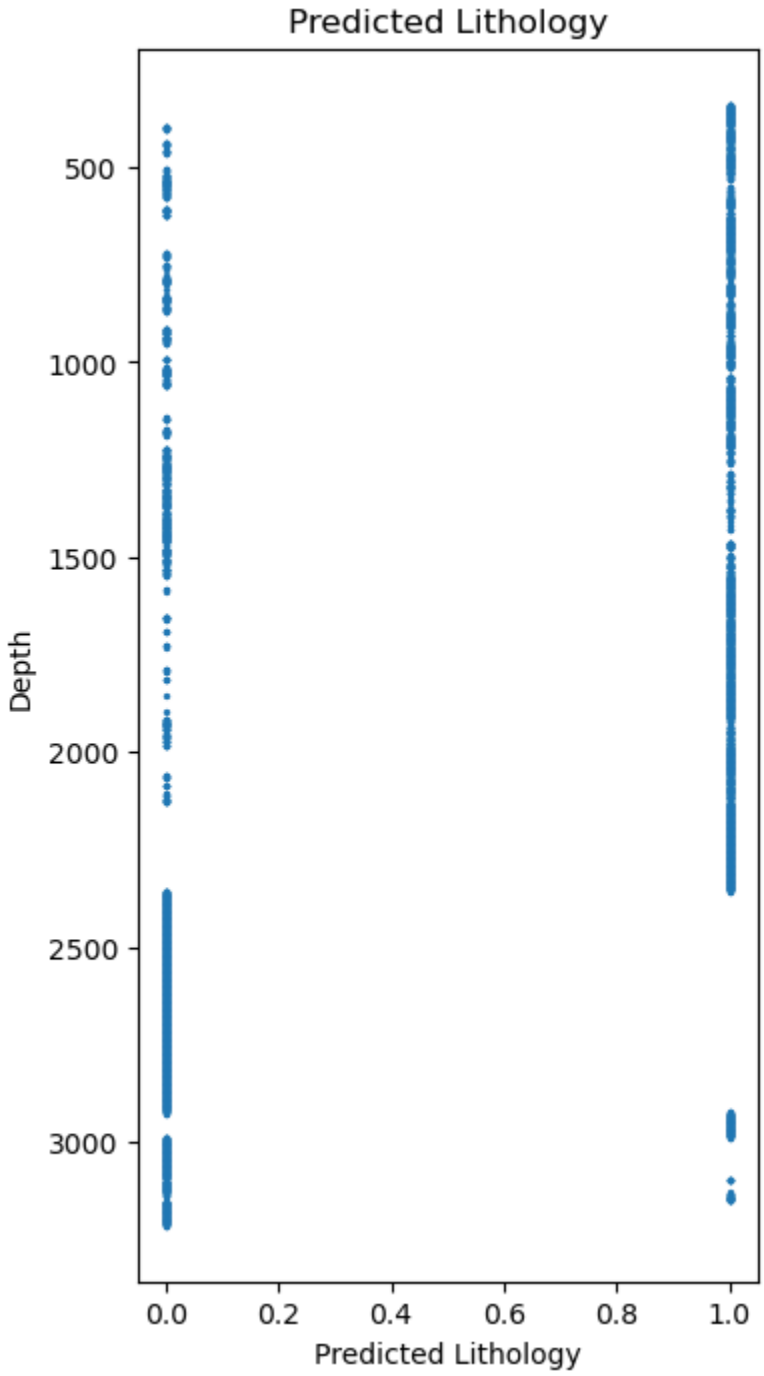

9. Interporetation and Conclusion

Machine learning classification was used to predict lithology from well log measurements. The Random Forest model was the algorithm selected due to its published woderful performance on geologic dataset. The developed model achieved strong predictive performance, with Gamma Ray emerging as the most influential feature. The predicted lithology intervals align well with conventional log interpretation, and consequently proves reliable.

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